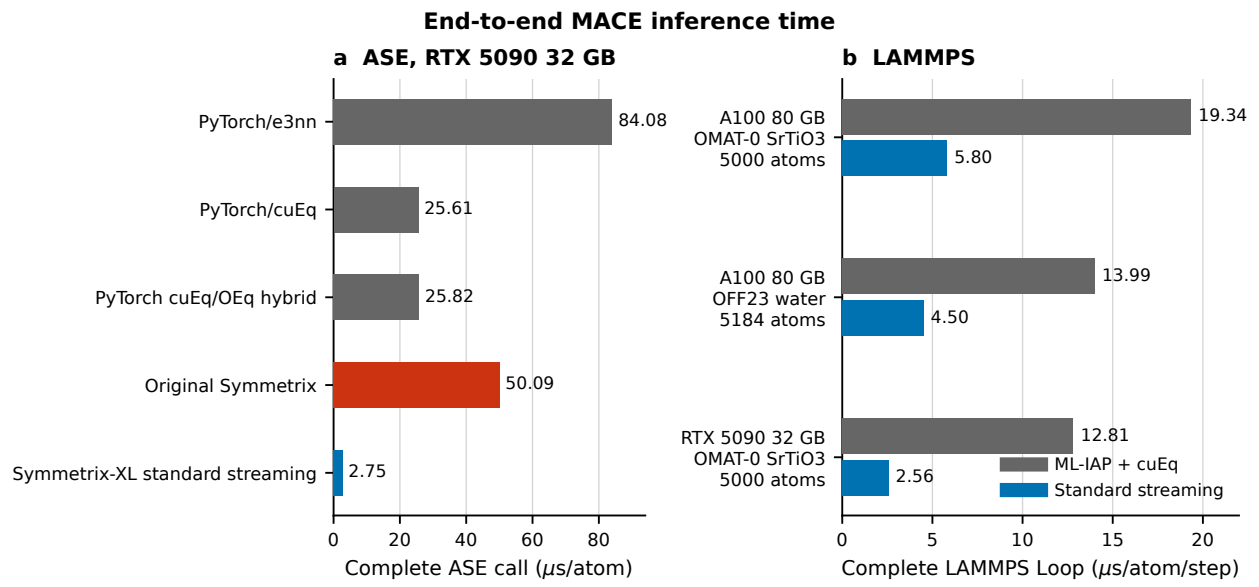

End-to-end MACE inference time
a ASE, RTX 5090 32 GB
PyTorch/e3nn
84.08
PyTorch/cuEq
25.61
PyTorch cuEq/OEq hybrid
25.82
Original Symmetrix
50.09
Symmetrix-XL standard streaming
2.75
0
20
40
60
80
Complete ASE call (μs/atom)
b LAMMPS
A100 80 GB
OMAT-0 SrTiO3
5000 atoms
19.34
5.80
A100 80 GB
OFF23 water
5184 atoms
13.99
4.50
RTX 5090 32 GB
OMAT-0 SrTiO3
5000 atoms
12.81
2.56
ML-IAP + cuEq
Standard streaming
0
5
10
15
20
Complete LAMMPS Loop (μs/atom/step)